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Comparative Analysis Of Deep Learning Approaches Used For Stock Price Prediction

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***Abstract*—Stock market analysis is nonlinear, highly volatile and complex. Predicting the stock price is a highly challenging task. The change in stock price is affected by certain factors such as economic situation, political circumstances, sentiments, global pandemic like Covid-19 and global impact of Russo-Ukrainian War. Forecasting the stock market presents a significant difficulty to investors to build a profitable portfolio or to reduce stock risk in the stock market. "Artificial Intelligence (AI), Machine Learning (ML) and Deep Learning (DL") have been abundantly used for stock trend and price prediction considering time series data and sentiments. In this paper we consider the stocks of 5 companies (Tata Steel, Tata Motors, Sun Pharmacy, Infosys and HDFC Bank) from National Stock Exchange (NIFTY 50). "Convolution Neural Network (CNN), Recurrent Neural Network (RNN), Long Short-Term Memory (LSTM) and Bidirectional Long Short-Term Memory (Bi-LSTM)" methods were used for data set of duration 01 January 2014 - 01 February 2024, was evaluated the prediction model using "Root Mean Squared Error (RMSE) and Mean Absolute Percentage Error (MAPE)". The Bi-LSTM model achieved an RMSE of 6.6 for TATA Steel stock and a MAPE of 0.010 for Sun Pharma stock. The study proves the Bi-LSTM is an effective deep learning algorithm for predicting the future closing price of a stock as compared to CNN, RNN and LSTM.**

Keywords— Stock Market, AI, ML, DL, CNN, RNN, LSTM and Bi-LSTM

I. Introduction

People invest their hard-earned money in the stock market to build a strong financial portfolio as compared to other financial investment tools. To protect their capital in an economic crisis is very important. Economic crises not only affect individuals but also affect the economy of the country. With the advancement of data analytics, artificial intelligence and improved computational abilities, forecasting the trend of the stock price is becoming possible. Earlier, it was difficult to figure out the movement of the stock price by manually reading the chaotic market data.

A high prediction accuracy rate offers the chance to build a successful portfolio. One of the most popular methods is the use of AI techniques to extract the stock trend from vast volumes of data, but as of now, there is hardly any algorithm or model that predicts the closing price of stock correctly. It is complicated and time-consuming work for an investor to take into account every factor e.g global sentiments or political news while selecting the stocks and predicting the stock price. Stock's next day closing price prediction needs historical data, market trend and technical indicators.

Stock Market Analysis include -

- "Fundamental Analysis".
- "Technical Analysis".
- "Sentiment Analysis".

"Fundamental Analysis (FA)"

It determines the true price of a stock. It identifies companies with strong fundamentals that are suitable for long-term investment.

"Technical Analysis (TA)"

Its focus lies on the stock market rather than the specific company. Using historical data of a stock, it forecasts the movement of its price.

"Sentiment Analysis (SA)"

The study that examined people's opinions and expressions, categorized as negative, positive or neutral is called "Sentiment Analysis".

II. Related Work

The prediction of stock's closing price is challenging task for researchers. Many researchers used deep learning methods for stock predictions, with different data sets, time frame and evaluation methods.

Y. Zhao et al. [1], proposed a new model SA-DLSTM, which incorporates investors sentiment, important features of stock's data and LSTM for prediction purpose. Their studies conclude that SA-DLSTM performs better than other conventional methods. Annual Return by SA-DLSTM model is 47.21 %.

J. Maqbool et al. [2], employed an MLP-Regressor model to examine data from HDFC, Reliance, Tata Motors, and Tata Steel. Their analysis yielded a notable prediction accuracy of 0.90 for a timeframe of 10 days.

In the analysis by M. Bansal et al. [4], comparing "Linear Regression (LR), Support Vector Machine (SVM), K Nearest Neighbor, Decision Tree Regression, and LSTM models" for NSE prediction, the LSTM algorithm demonstrated superior performance. Particularly, it yielded the lowest MAPE (1.59),

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RMSE (22.55) and R2 (-0.11) values, indicating its heightened accuracy in this domain.

S. Albahli et al. [5], used the "S&P 500 dataset" to study the predictive capabilities of the DenseNet-41 model. Their research yielded impressive results, achieving a minimum MAPE score of 0.32.

X. Liu et al. [6] utilized the ISSA-BP neural network model to forecast various stock indices such as SSE, SZSE, S&P500 and DJI. Their results showcased promising predictive capabilities, with a MAPE (%) of 0.1666 and RMSE of 10.1330 for SSE and a MAPE (%) of 0.0934 and RMSE of 3.1980 for S&P 500, affirming the effectiveness of the "ISSA-BP neural network" used to predict the stock market.

In the investigation by A. Gangwar et al. [7] on LSTM and ANN models for S&P500 index prediction, the ANN model emerged as the most proficient. With the lowest "MAPE (0.77), RMSE (0.42) and MBE (0.013)" values, the ANN model demonstrated superior accuracy in forecasting the S&P 500 index.

P. Chhajer et al. [8], presents a broad review of the use of AI and ML in stock market predictive analytics. Data sets referred in various review papers are S&P 500, Nifty 50 and Nikkei 225. Review papers states that LSTM model performs better than ANN and SVM.

J. Shen et al. [11], suggested approach using "deep learning and feature engineering" for prediction of the trend in the price of stock. Chinese stock market data from 2017 to 2018 is used It uses a feature engineering process along with an LSTM prediction model. LSTM trained on 29 selected features, Model Performance Metric score –F1 score is 0.93, Precision is 0.96, Recall is 0.96 & Accuracy is 0.93

In K. Zhang et al.'s study [12], which explored the application of GAN and LSTM models for predicting stock indices including S&P 500, NYSE, and NASDAQ, the GAN model exhibited enhanced performance with low MAE (3.0401), MAPE (0.0137) and RMSE (4.1026) values.

In the research conducted by Hiransha M et al. [14], numerous conventional time series model such as "AR, MA, Auto Regressive Moving Average (ARMA), Auto Regressive Integrated Moving Average (ARIMA), Autoregressive Conditional Heteroskedasticity (ARCH) and Generalized Autoregressive Conditional Heteroskedasticity (GARCH)" were compared against several DL models including "ANN, MLP, RNN and LSTM" for predicting stock indices such as NSE and NYSE. The outcomes of research showed that the DL models consistently outperformed the ARIMA model, indicating the superior predictive abilities of deep learning approaches in this domain.

R. Ren et al. [15], they use sentiment features that take into account investor psychology in addition to financial market data to identify the trend of stock market. Data set of SSE 50 companies of for time period (17/06/2014 – 7/06/2016) were used. Automatic feature extraction from unstructured news data results in the expression of the features as sentiment indices. They integrate SVM for sentiment analysis. They achieved 89.93% forecasting accuracy after integrating sentiment variables.

Many researchers developed hybrid models for stock trend prediction and stock price prediction. Mostly working on developing a model which incorporates sentiments with the historical data of a specific company.

This research paper provides a comparative analysis of various deep learning algorithms used to predict stock market trends.

III. Implementation

In this research, the data set of five companies (Tata Steel, Tata Motors, Sun Pharmacy, Infosys and HDFC Bank) is used. This study considers the time frame of 10 years- 1 January 2014 to 1 January 2024 [16]. We have taken data from yahoo finance website.

Stock market data of includes features like "opening price, highest price, lowest price, closing price, transaction date and volume". We consider closing price of stock for study.

We have implemented the prediction model using "CNN, RNN, LSTM and Bi-LSTM" methods.

The model is developed in the "Python programming language using Jupyter Notebook version: 5.3.0".

TensorFlow version: 2.15.0

Hardware Specifications -

Architecture: ('64bit', 'Windows PE'), System: Windows, RAM: 8GB, "Processor: Intel64 Family 6 Model 61 Stepping 4"

A. CNN Model

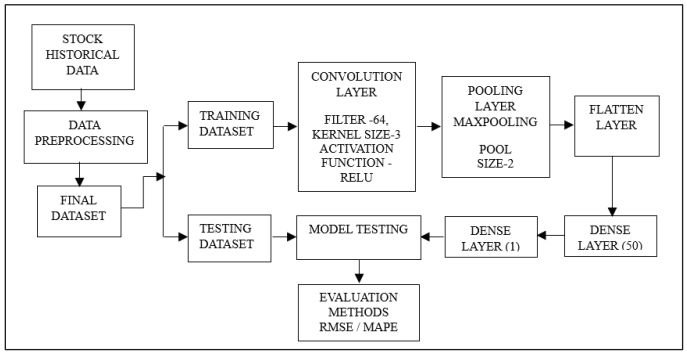

Fig. 1. Implemented CNN Prediction Model

As shown in Fig 1, One-dimensional convolutional layer is incorporated, containing 64 filters and size of kernel was three. The "Rectified Linear Unit (ReLU)" activation function is used to introduce nonlinearity. Subsequently, a "max-pooling layer" with a pool size of two is utilized to reduce the spatial dimensions of the convolutional output while emphasizing essential characteristics. Following this, a "flatten layer" is introduced to convert the pooled feature maps into a one-dimensional vector, preparing them for input into the subsequent fully connected layers. Two "dense layers" is used, the first comprising 50 neurons activated by ReLU to facilitate nonlinear transformations and feature extraction, and the final layer with one neuron tailored for regression tasks, delivering output

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predictions. The model is developed using the "Adam optimizer", which is well-known for its efficiency in training deep neural networks. Training lasts "100 epochs with a batch size of 64", ensuring regular weight adjustments to improve the model's predictive performance. "Root Mean Square Error (RMSE) and Mean Absolute Percentage Error (MAPE)" are used as evaluation methods. For all other model "100 epochs and a batch size of 64" are employed. "RMSE and MAPE" evaluation methods are used for all other models.

B. RNN Model

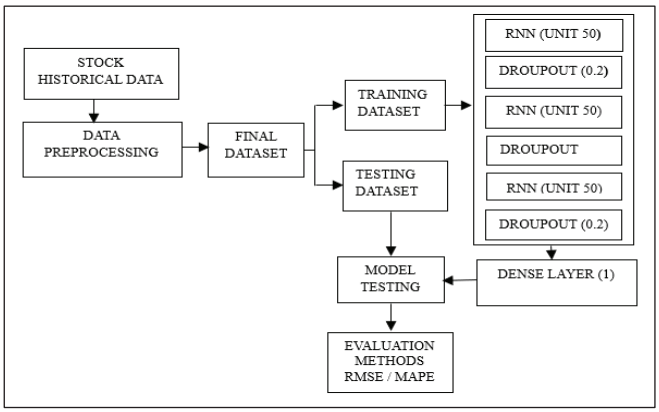

Fig. 2. Fig. 2. Implemented RNN Prediction Model

As shown in Fig 2, a sequential neural network model is implemented. It begins by adding a RNN layer of 50 units. Dropout regularization is then used to prevent overfitting, with a 0.2 dropout rate. Subsequent layers follow a similar pattern: each RNN layer contains 50 units and Similar to the first two layers, it is intended to return sequences, followed by a dropout layer for regularization. Finally, the output layer is a tightly coupled layer with a single unit, which is designed for regression tasks. Overall, this architecture uses recurrent neural networks (RNNs) to process sequential input, with dropout regularization to improve generalization and a single-unit output layer for prediction.

C. LSTM Model

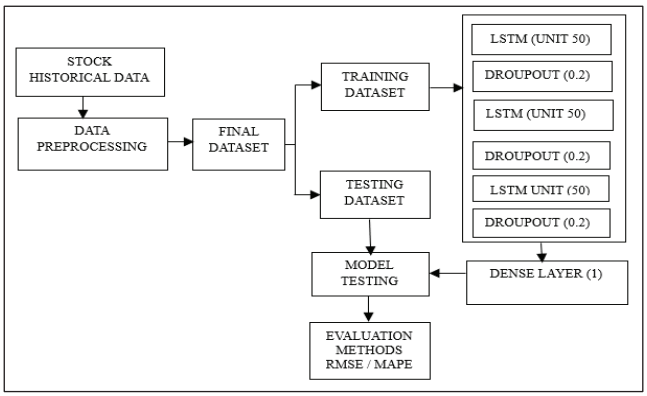

Fig. 3. Implemented LSTM Prediction Model

Fig 3 shows the implemented LSTM model, the model architecture consists of three "Long Short-Term Memory (LSTM)" layers, each with 50 neurons, to capture complex temporal patterns. To minimize overfitting, "dropout layers" with a 0.2 dropout rate are added between the "LSTM layers". A "dropout layer" is placed after each "LSTM layer" to make the network more consistent.

D. Bi-LSTM Model

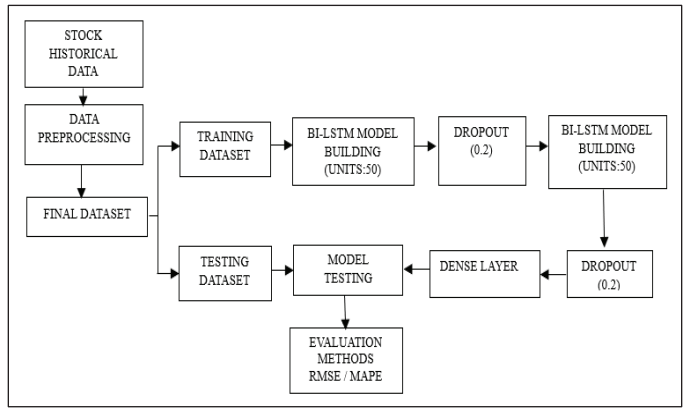

Fig. 4. Implemented Bi-LSTM Prediction Model

The code creates a sequential model with a linear stack of layers. Fig 4 shows the implemented Bi-LSTM model. It includes a Bidirectional LSTM layer with 50 units that is designed to record both the previous and future context of sequence, and it anticipates input sequences formed according to the training data. A "dropout layer" is used to prevent overfitting by randomly deactivating 20% of the input units during training. Finally, an output layer is added, which is densely coupled and has one unit.

IV. Result and Discussions

This study considers the stocks Infosys, Tata Motors, HDFC Bank, Tata Steel and Sun Pharma of NIFTY 50. These all stocks are major contributor in their sectorial indices. RMSE and MAPE are used as evaluation tools.

A. Root Mean Squared Error

RMSE is often used metric in stock market forecasting because of its ability to quantify prediction accuracy, sensitivity to errors, interpretability, comparability and suitability for optimization.

RMSE is calculated as-

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=0}^{n}(yi - y\hat{}i)^2} \quad (1)$$

where -n denotes the number of observations.

yi -denotes the variable's actual value at the ith observation and y^i denotes the expected value of a variable at the ith observation.

B. Mean Absolute Percentage Error

MAPE shows the average percentage error in predictions relative to the actual values. It is often used in predicting the accuracy of models, especially when dealing with different scales of data. A lower MAPE indicates better accuracy.

"Mean Absolute Percentage Error" is computed as-

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$$MAPE = \frac{1}{n} \sum_{i=0}^{n} \frac{(yi - y\hat{}i)}{yi} \quad (2)$$

Table 1 given below, represents the RMSE findings for the "CNN, RNN, LSTM and Bi-LSTM models" applied to dataset of Tata Steel, Tata Motors, Sun Pharma, Infosys and HDFC Bank.

TABLE I. EVALUATION METHOD- RMSE

Name of Stocks (NSE	Time Frame (01 January 2014 - 01 February 2024)				
	Models	CNN	RNN	LSTM	Bi-LSTM
	Evaluation Parameters	RMSE	RMSE	RMSE	RMSE
TATA STEEL (TATASTEEL.NS)		10.37	5.58	7.57	6.06
TATA MOTORS (TATAMOTORS.NS)		17.55	14.71	15.52	13.02
SUNPHARMA (SUNPHARMA.NS)		14.85	17.12	23.65	14.49
INFOSYS (INFY.NS)		90.03	59.54	45.04	42.07
HDFC BANK (HDFCBANK.NS)		48.80	40.34	49.37	31.19

The findings indicates that the Bi-LSTM model yielded the lowest RMSE across all stocks compared to other models.

Table 2 given below, represents the MAPE findings for the "CNN, RNN, LSTM and Bi-LSTM models" applied to dataset of Tata Steel, Tata Motors, Sun Pharma, Infosys and HDFC Bank

TABLE II. EVALUATION METHOD- MAPE

Name of Stocks (NSE	Time Frame (01 January 2014 - 01 February 2024)				
	Models	CNN	RNN	LSTM	Bi-LSTM
	Evaluation Parameters	MAPE	MAPE	MAPE	MAPE
TATA STEEL (TATASTEEL.NS)		0.079	0.038	0.033	0.031
TATA MOTORS (TATAMOTORS.NS)		0.025	0.025	0.024	0.021
SUNPHARMA (SUNPHARMA.NS)		0.012	0.013	0.019	0.010
INFOSYS (INFY.NS)		0.051	0.029	0.023	0.021
HDFC BANK (HDFCBANK.NS)		0.027	0.021	0.027	0.015

The results proves that the lowest MAPE is achieved using Bi-LSTM model than CNN, RNN and LSTM model.

Here are the visual representations illustrating the actual and forecasted values of Tata Steel stock across the CNN, RNN, LSTM and Bi-LSTM models.

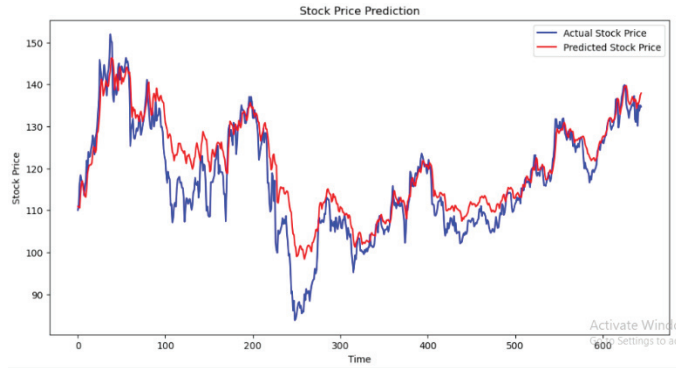

Fig. 5. Plot of Tata Steel CNN model

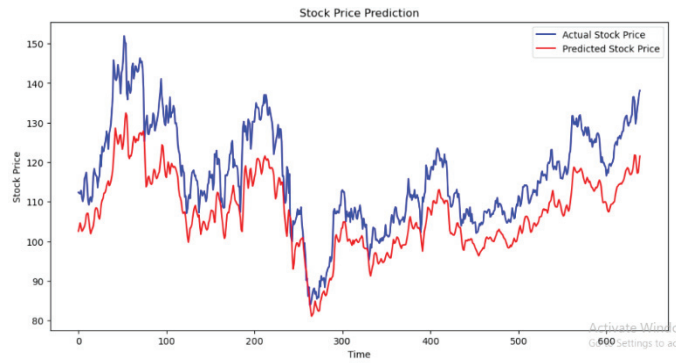

Fig. 6. Plot of Tata Steel RNN model

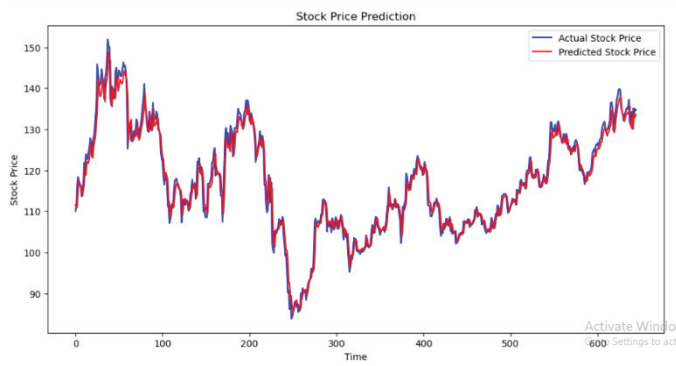

Fig. 7. Plot of Tata Steel LSTM model

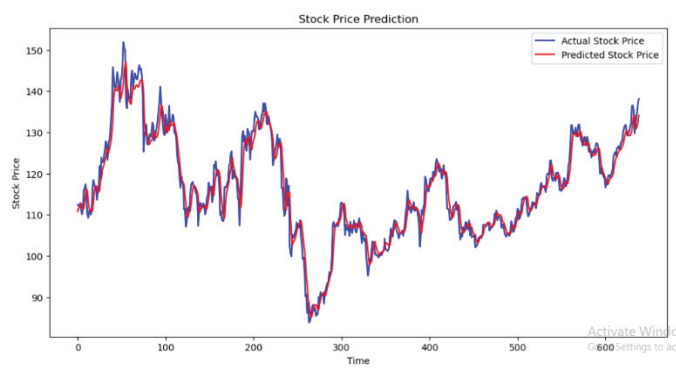

Fig. 8. Plot of Tata Steel Bi-LSTM model

Among all the charts depicting Tata Steel stock data, the Bi-LSTM model demonstrates superior performance, particularly

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in scenarios where capturing bidirectional temporal dependencies is crucial.

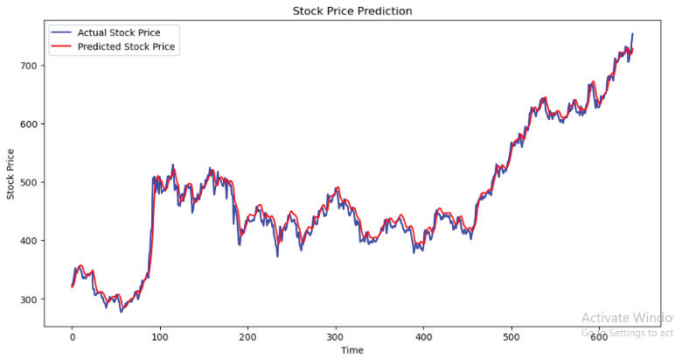

Fig. 9. Plot of Tata Motors Bi-LSTM model

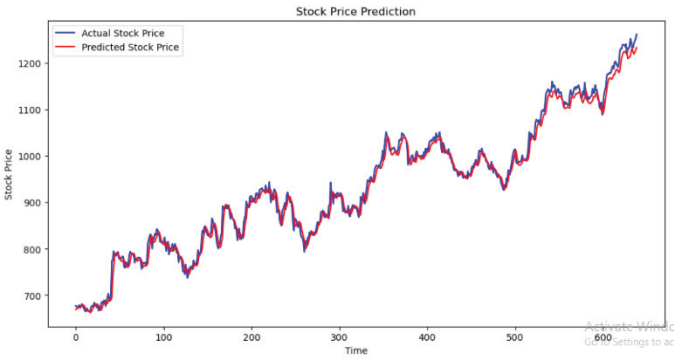

Fig. 10. Plot of Sun Pharma Bi-LSTM model

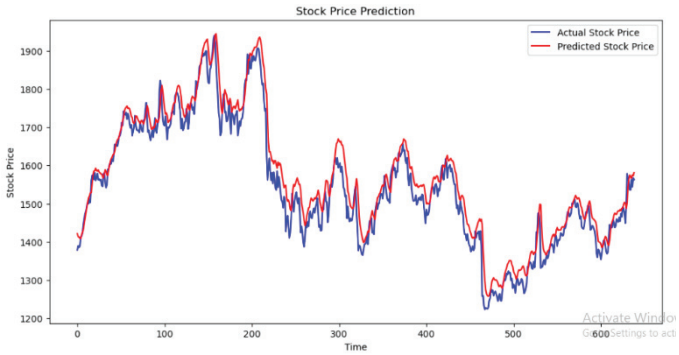

Fig. 11. Plot of Infosys Bi-LSTM model

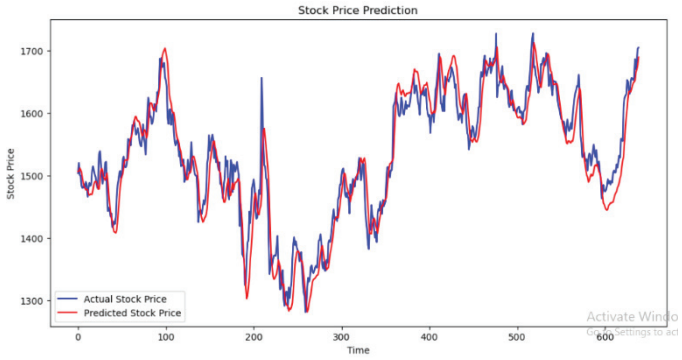

Fig. 12. Plot of HDFC Bank Bi-LSTM model

Fig No. 9, 10,11 and 12 shows the prediction results of Tata Motors, Sun Pharma, Infosys and HDFC bank using Bi-LSTM model. Bi-LSTM is often preferred over "CNN, RNN and LSTM" in "Stock market prediction" due to its capability to capture bidirectional dependencies in sequential data, allowing it to incorporate both past and future context effectively. This bidirectional processing allows Bi-LSTM to better describe the complex temporal patterns found in stock market data, perhaps leading to higher forecast accuracy than other methods.

V. Conclusion

In this study, five stocks Infosys, Tata Motors, HDFC Bank, Tata Steel and Sun Pharma from the Nifty 50 index were scrutinized using predictive models including CNN, RNN, LSTM and Bi-LSTM, with RMSE and MAPE employed for evaluation. This study considers a timeframe from January 1, 2014, to January 1, 2024, encompassing the COVID-19 pandemic period, where significant fluctuations are accounted for, enhancing the robustness of the implementation. The Bi-LSTM model achieved an RMSE of 6.6 for TATA Steel stock and a MAPE of 0.010 for Sun Pharma stock. These results suggest that the Bi-LSTM model demonstrates promising predictive accuracy for both stocks, with relatively low error metrics indicating its effectiveness in forecasting their respective closing prices.

VI. Limitations and Future Scope

Stock market is sentiment driven, so stock price of a specific company is highly affected by the news related to that companies. Sentiment analysis can be incorporated with the historical data for better prediction accuracy. Hybrid models significantly improve the predictability of stocks closing prices.

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